Questions

Q1.

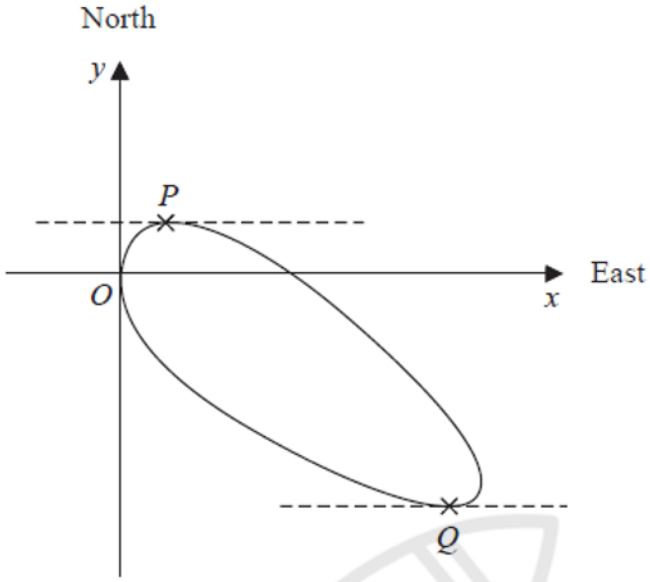


Figure 4

Figure 4 shows a sketch of the closed curve with equation

$$(x+y)^3 + 10y^2 = 108x$$

(a) Show that

$$\frac{dy}{dx} = \frac{108 - 3(x+y)^2}{20y + 3(x+y)^2}$$

(5)

The curve is used to model the shape of a cycle track with both x and y measured in km.

The points P and Q represent points that are furthest north and furthest south of the origin O, as shown in Figure 4.

Using the result given in part (a),

(b) find how far the point Q is south of O. Give your answer to the nearest 100 m.

(4)

(Total for question = 9 marks)

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(Q11 WMA14/01, Oct 2022)

Q2.

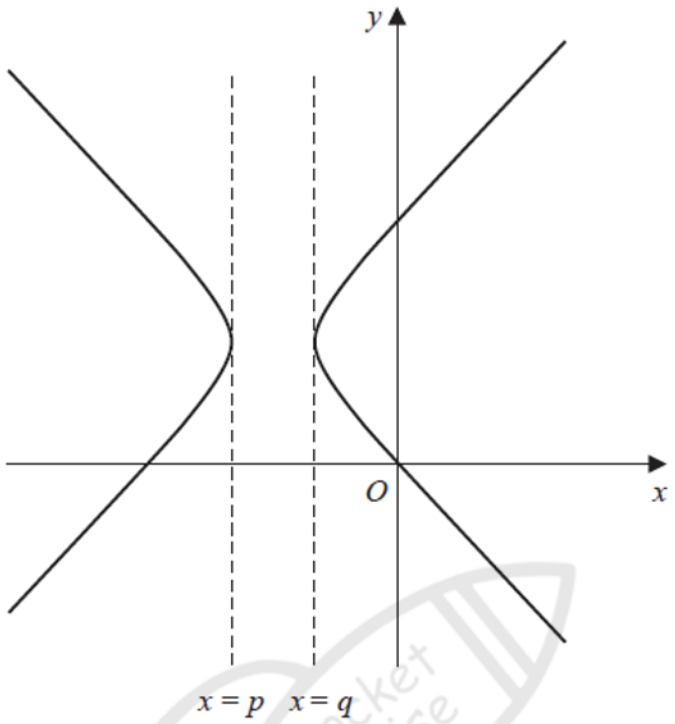


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y^2 = 2x^2 + 15x + 10y$$

(a) Find $\frac{dy}{dx}$ in terms of x and y.

(4)

The curve is not defined for values of x in the interval (p, q), as shown in Figure 2.

(b) Using your answer to part (a) or otherwise, find the value of p and the value of q.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(3)

(Total for question = 7 marks)

(Q05 WMA14/01, Jan 2023)

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Q3.

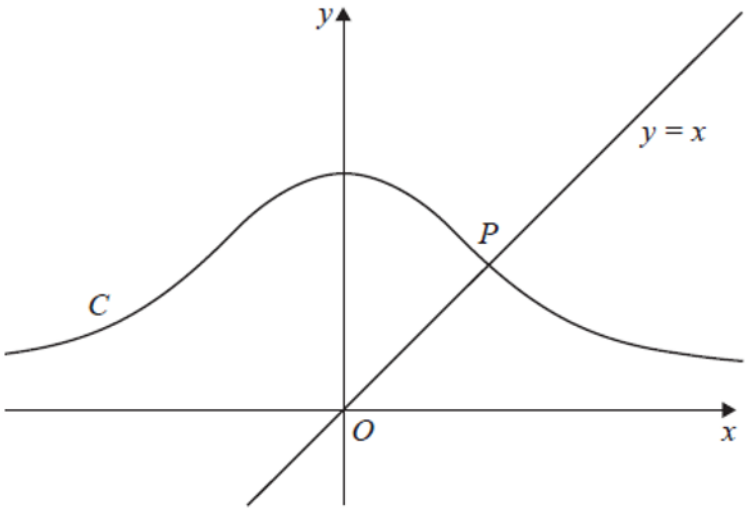


Figure 2

Figure 2 shows a sketch of the curve C with equation

$$y^3 - x^2 + 4x^2y = k$$

where k is a positive constant greater than 1

(a) Find $\frac{dy}{dx}$ in terms of x and y.

(5)

The point P lies on C.

Given that the normal to C at P has equation $y = x$, as shown in Figure 2,

(b) find the value of k.

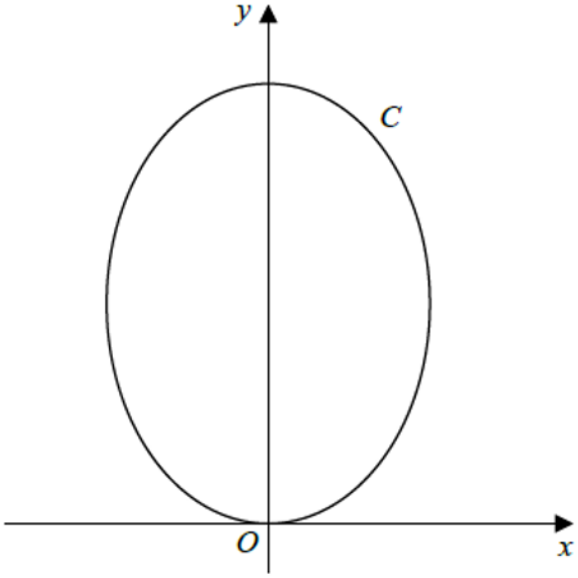
(5)

(Total for question = 10 marks)

(Q05 WMA14/01, Oct 2023)

Q4.

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Figure 1

Figure 1 shows a sketch of the curve C with parametric equations

$$x = \sqrt{3}\sin 2t \qquad y = 4\cos^2 t \qquad 0 \leqslant t \leqslant \pi$$

(a) Show that $\frac{dy}{dx} = k\sqrt{3}\tan 2t$, where k is a constant to be found.

(5)

(b) Find an equation of the tangent to C at the point where $t = \frac{\pi}{3}$

Give your answer in the form $y = ax + b$, where a and b are constants.

(4)

(Total for question = 9 marks)

(Q04 WMA14/01, Specimen papers)

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Q5.

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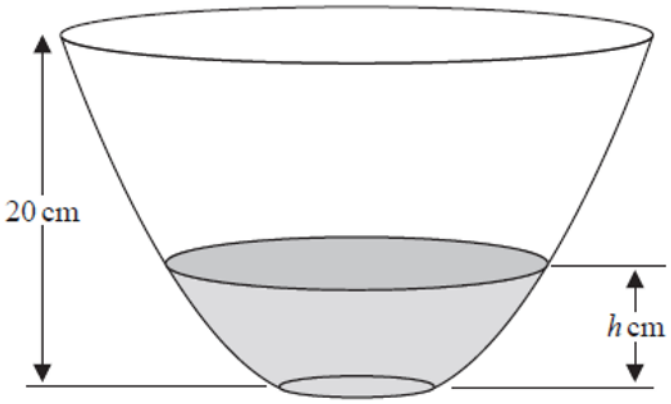


Figure 2

A bowl with circular cross section and height 20 cm is shown in Figure 2.

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The bowl is initially empty and water starts flowing into the bowl.

When the depth of water is h cm, the volume of water in the bowl, V cm^3, is modelled by the equation

$$V = \frac{1}{3} h^2 (h + 4) \qquad 0 \leq h \leq 20$$

Given that the water flows into the bowl at a constant rate of 160 cm^3 s^{-1}, find, according to the model,

(a) the time taken to fill the bowl,

(2)

(b) the rate of change of the depth of the water, in cm s^{-1}, when $h = 5$

(5)

(Total for question = 7 marks)

(**Q03 WMA14/01, June 2021**)

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Q6.

The curve C has equation

$$2x - 4y^2 + 3x^2 y = 4x^2 + 8$$

The point $P(3, 2)$ lies on C.

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Find the equation of the normal to C at the point P, writing your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(7)

(Total for question = 7 marks)

(Q01 WMA14/01, Oct 2021)

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Q7.

A curve C has equation

$$y = x^{\sin x} \qquad x > 0 \qquad y > 0$$

(a) Find, by firstly taking natural logarithms, an expression for $\frac{dy}{dx}$ in terms of x and y.

(5)

(b) Hence show that the x coordinates of the stationary points of C are solutions of the equation

$$\tan x + x \ln x = 0$$

(2)

(Total for question = 7 marks)

(Q06 WMA14/01, Oct 2020)

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Q8.

A curve C has the equation

$$x^3 + 2xy - x - y^3 - 20 = 0$$

(a) Find $\frac{dy}{dx}$ in terms of x and y.

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(5)

(b) Find an equation of the tangent to C at the point $(3, -2)$, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(2)

(Total for question = 7 marks)

(Q02 WMA14/01, Specimen papers)

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Q9.

A curve has equation

$$4y^2 + 3x = 6ye^{-2x}$$

(a) Find $\frac{dy}{dx}$ in terms of x and y.

The curve crosses the y-axis at the origin and at the point P.

(b) Find the equation of the normal to the curve at P, writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(4)

(Total for question = 4 marks)

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(Q06 WMA14/01, Jan 2021)

Q10.

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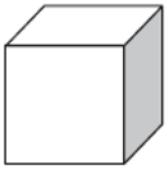


Figure 1

Figure 1 shows a cube which is increasing in size.

At time t seconds,

- the length of each edge of the cube is x cm
- the surface area of the cube is S cm^2
- the volume of the cube is V cm^3

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Given that the surface area of the cube is increasing at a constant rate of 4 cm^2 s^{-1}

(a) show that $\frac{dx}{dt} = \frac{k}{x}$ where k is a constant to be found,

(4)

(b) show that $\frac{dV}{dt} = V^p$ where p is a constant to be found.

(3)

(Total for question = 7 marks)

(**Q02 WMA14/01, Oct 2023**)

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Q11.

A curve has equation

$$y^2 = y\,e^{-2x} - 3x$$

(a) Show that

$$\frac{dy}{dx} = \frac{2ye^{-2x} + 3}{e^{-2x} - 2y}$$

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(4)

The curve crosses the y-axis at the origin and at the point P.

The tangent to the curve at the origin and the tangent to the curve at P meet at the point R.

(b) Find the coordinates of R.

(5)

(Total for question = 9 marks)

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(Q05 WMA14/01, June 2021)

Q12.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$16x^3 - 9kx^2y + 8y^3 = 875$$

where k is a constant.

(a) Show that

$$\frac{dy}{dx} = \frac{6kxy - 16x^2}{8y^2 - 3kx^2}$$

(4)

Given that the curve has a turning point at $x = \frac{5}{2}$

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(b) find the value of k

(4)

(Total for question = 8 marks)

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(Q04 WMA14/01, June 2022)

Q13.

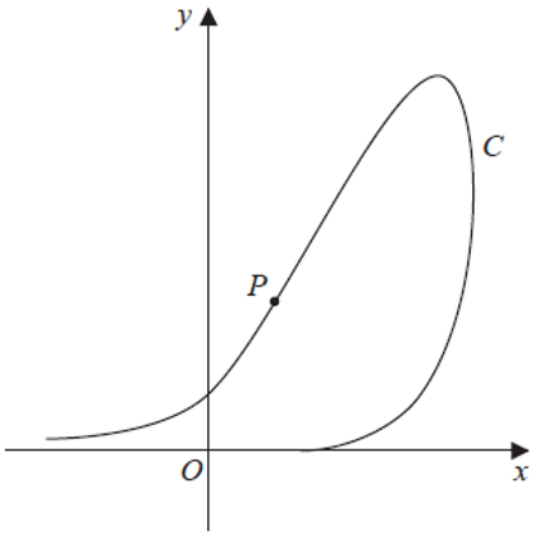


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$2x - 4xy + y^2 = 13 \qquad y \geq 0$$

The point P lies on C and has x coordinate 2

(2)

(b) Find $\frac{dy}{dx}$ in terms of x and y.

(5)

The tangent to C at P crosses the x-axis at the point Q.

(c) Find the x coordinate of Q, giving your answer in the form $\frac{a\ln 2 + b}{c\ln 2 + d}$ where a, b, c and d are integers to be found.

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(3)

(Total for question = 10 marks)

(Q02 WMA14/01, June 2023)

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Q14.

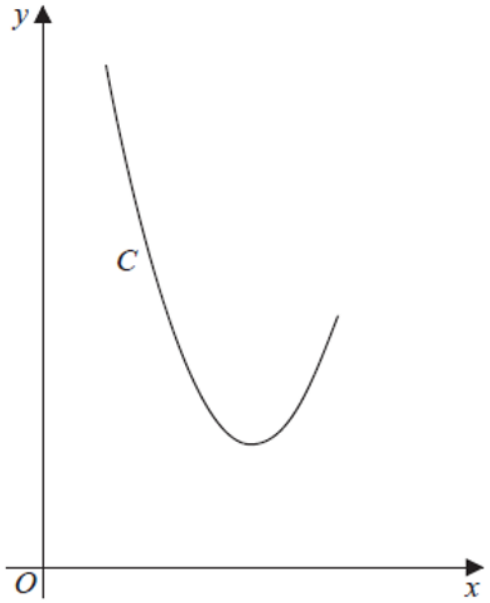


Figure 1

Figure 1 shows a sketch of the curve C with parametric equations

$$x = 5 + 2\tan t \qquad y = 8\sec^2 t \qquad -\frac{\pi}{3} \leqslant t \leqslant \frac{\pi}{4}$$

(a) Use parametric differentiation to find the gradient of C at $x = 3$

(4)

The curve C has equation $y = \mathrm{f}(x)$, where f is a quadratic function.

(b) Find $\mathrm{f}(x)$ in the form $a(x + b)^2 + c$, where a, b and c are constants to be found.

(3)

(c) Find the range of f.

(2)

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(Total for question = 9 marks)

(Q05 WMA14/01, Oct 2021)